Finitary derivability is compact: proofs use finitely many assumptions

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: this is syntactic compactness, prior to and independent of the completeness theorem, as the section preamble says; the enumerate layout with both items is kept verbatim.

Editorial note. The label prop:proves-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Definition of derivation. Earlier in the same chapter: derivations are finite sequences of formulas, each an axiom, an assumption, or a rule consequence. Line 918: the finiteness of the sequence is the whole engine of item (1).
- Inconsistency as derivation of falsum. Section opening definitions; \ollabel{prop:incons} states the equivalence (its proof is the author's exercise). Item (2) is instantiated at A = falsum at line 925.

Proposition 0.1 (Compactness)**.** *1. If* $\Gamma \vdash !A$ *then there is a finite subset* $\Gamma_0 \subseteq \Gamma$ *such that* $\Gamma_0 \vdash !A$*.*

2. If every finite subset of Γ *is consistent, then* Γ *is consistent.*

Proof. 1. If $\Gamma \vdash !A$, then there is a finite sequence of !!formulas $!A_1$, …, $!A_n$ so that $!A \,\mathrm{id}\, !A_n$ and each $!A_i$ is either a logical axiom, !!aelement of Γ or follows from previous !!formulas by modus ponens. Take Γ_0 to be those $!A_i$ which are in Γ. Then the !!derivation is likewise !!aderivation from Γ_0, and so $\Gamma_0 \vdash !A$.

2. This is the contrapositive of (1) for the special case $!A \,\mathrm{id}\, \bot$.

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